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**On calibrated inverse probability weighting
and generalized boosting propensity score
models for mean estimation with incomplete
survey data**

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For Peer Review

Abstract

Incomplete data, whether realized from nonresponse in survey data or counterfactual outcomes in observational studies, may lead to biased estimation of study variables. Nonresponse and selection bias may be mitigated with techniques that weight the incomplete data to match characteristics of the partially unobserved complete data. Inverse probability weighting is a widely used method in causal inference that relies on a propensity model to construct adjusted weights; whereas calibration is a common method used by survey statisticians to use constrained optimization to construct adjusted weights. This paper reviews inverse probability weighting and a particular calibration method by distinguishing them in the statistical sense of variable balancing, extending propensity score construction to include generalized boosting models, and demonstrating the use of inverse probability weighting and calibration separately and together through a widely cited simulation study evaluation.

1 Introduction

Mean estimation may be biased in the presence of incomplete data. For example, unit nonresponse in survey data may lead to biased estimation when the respondents are not representative of the entire sample, and mean estimation from observational studies may be biased when treatment assignment is not random. There are a variety of statistical methods to mitigate incomplete data by adjusting observed data to reflect the complete data via weighting. Weight-based adjustments are well studied in survey research – e.g. calibration weighting methods (Wu and Lu, 2016), including the widely used raking method (Deming and Stephan, 1940) – and causal inference – e.g. inverse of probability weighting (Austin, 2011; Robins et al., 2000).

The use of calibration in survey statistics has a long history for the purpose of benchmarking. Survey statisticians have used various calibration weighting methods, including iterative proportional fitting procedure (Deming and Stephan, 1940), generalized raking procedures (Deville and Särndal, 1992), model-based calibration (Wu and Sitter, 2001), instrumental variable-related calibration (Kott, 2006) and the empirical likelihood approach to calibration (Wu and Rao, 2006). Wu and Lu (2016) discussed an extensive list of calibration methods, which all aim to approximate target totals of a population of interest. Beyond its use for achieving sample balance with known population values, survey calibration it may also be used in the context of missing data. That is, it has been used to adjust for nonresponse bias by calibrating respondent data to balance with the characteristics all sampling units: see, for example, Lundström and Särndal (1999); Chang and Kott (2008); Särndal (2008). Calibration has also been used to infer causal effects in observational studies; in particular, the entropy balancing weighting (EBW) calibration method has been amply cited in the field of political science (Hainmueller, 2012). Calibration motivated through causal inference aims to balance characteristics among comparison groups in observational studies, specifically treatment versus control. This context aligns with the use for nonresponse bias adjustment in surveys, where the comparison groups are respondents versus nonrespondents.

The initial development of causal inference methods was not based on calibration methods but instead on the propensity score (Rosenbaum and Rubin, 1983). Various propensity score-based methods have been widely used in economics (Abadie and Imbens, 2016), behavioral health (Austin, 2011), and social sciences (Thoemmes and Kim, 2011). A standard textbook in epidemiology (Hernán and Robins, 2020) describes propensity score-based inverse probability weighting (IPW) as a way of conducting causal analysis. IPW involves estimating a propensity (e.g. to be given a treatment, or to respond to a survey), the inverse of which is used to weight the observed group to match characteristics of the unobserved group.

We describe and assess two developments of calibration and IPW: (1) the use of a generalized boosted machine learning model to estimate the propensity in the IPW method, and (2) a calibrated IPW approach that combines the two weight adjustment methods. We assume a survey context in that we illustrate bias reduction in mean estimation of an outcome of interest through an estimate that may only use outcome information from observed (e.g. respondent) data. These two developments also naturally extend to the causal inference context, in which case evaluation would focus on quantities such as average causal effect.

The IPW method relies heavily on the correctness of the propensity score. Estimation of the propensity (e.g. to be treated or to respond to a survey) is often done using classic statistical models such as logistic regression, but any classification method is applicable. For example, machine learning models have been also used, including the generalized boosted regression model (Friedman, 2001; Hastie et al., 2001; Ridgeway, 2020). Machine learning classification algorithms may allow more flexibility such as through potentially deep interactions and complex propensity characteristic patterns. In the context of survey research, tree-based machine learning models have been studied and compared for modeling and correcting for survey nonresponse: see, for example, Toth and Phipps (2014); Lohr et al. (2015); Buskirk and Kolenikov (2015); Kern et al. (2019), and references therein. In the context of causal inference, Lee et al. (2010) compared the logistic regression, classification trees (CART), bootstrap aggregated (bagged) CART, random forests, and the generalized boosted model (GBM) to evaluate performances of these machine learning-based propensity scores. They reported that GBM provided consistently excellent performance in propensity score model building as the efficiency of boosting as a general algorithm to improve estimates is well-known in the machine learning world (Hastie et al., 2001). Aligned with findings by Lee et al. (2010), in this paper we review GBM and compare it to logistic regression for estimating propensity.

Both calibration and IPW share the same purpose of weighting an incomplete data set to make it look like a target population. With this conceptual similarity, it is natural and straightforward to combine these two methods, i.e. one can calibrate the IPW weights. Mercer et al. (2018) compare a specific calibration technique (raking), IPW, and combined methods with benchmarks drawn from high quality surveys covering a range of topics including demographic and political variables. They find similar absolute differences between population benchmarks and weighted sample estimates for all methods, with the method that

combined raking and IPW performing slightly better. They concluded that the combined method resulted in consistent improvement compared with raking alone.

Through reviewing and assessing the developments of machine learning-based propensity and calibrated IPW, this paper aims to (1) contribute an explanation of differences between calibration weighting and IPW in a statistical sense – from the perspective of balancing conditions – and (2) demonstrate better performances of the calibrated IPW through a simulation study routinely used for causal inference and missing data analysis (Kang and Schafer, 2007). The rest of this paper proceeds as follows: Section 2 reviews the IPW method, Section 3 reviews calibration and its use to ensure the balancing condition of the IPW method, Section 4 demonstrates superior performances of the calibrated IPW method over calibration weighting only and the IPW method only, and Section 5 concludes.

2 Review of inverse probability weighting (IPW)

2.1 The IPW formulation

Suppose we have the following random variables: Y a continuous outcome, R the indicator for the missingness of Y ($R=1$ if observed), and $X = (X_1, \dots, X_p)$ a set of covariates for a data set of size N . Throughout this paper, random variables and their realized values are uppercase and lowercase letters, respectively. In addition, let π denote the true conditional probability of R given X , i.e. $\pi = P(R|X)$.

Table 1: Data structure for IPW formulation

i	R	X_1	...	X_p	Y
1	1	x_{11}	...	$x_{p,1}$	y_1
2	1	x_{12}	...	$x_{p,2}$	y_2
...	...	...	...	...	...
n	1	x_{1n}	...	$x_{p,n}$	y_n
$n + 1$	0	$x_{1,n+1}$	...	$x_{p,n+1}$	?
...	...	...	...	...	...
N	0	x_{1N}	...	$x_{p,N}$	?

Table 1 displays the components of the data structure assumed in this paper. The question marks in Table 1 are the missing values in Y for units where $R = 0$. True values of Y always exist, but values of Y are observed only when $R = 1$. We assume that X is fully observed.

If the mechanism π does not depend on X , as in a randomized experiment, then $E(Y)$ could easily be estimated by the sample mean of Y among $R = 1$. But when π does depend on X , as it is assumed in this paper, then $E(Y)$ will be biased. The driving motivation in IPW is to weight the multivariate covariate probability distribution for the observed data $P(X|R = 1)$

to make it approximate the multivariate covariate probability distribution for all the data $P(X)$. Ridgeway et al. (2015) and Kang et al. (2020) used this notion of balancing in causal inference for survey-weighted data and latent class exposures, respectively. The following definition clarifies the notion by weights w .

Definition 1. *The multivariate balancing condition for w is defined as*

$$w \times P(X|R = 1) = P(X). \quad (1)$$

That is, there is a w that can adjust the multivariate covariate distribution for the observed data to equal the multivariate covariate distribution for the full data. It can be shown that by the Bayes rule, equation (1) becomes

$$w \approx \frac{1}{P(R = 1|X)}, \quad (2)$$

where $P(R = 1|X)$ is the true conditional probability for R given X , which is known as the propensity score (Rosenbaum and Rubin, 1983). The propensity score is a concise, scalar summary of the propensity as a function of covariates X . Note that definition (1) requires the true propensity score in constructing the IPW weights in equation (2). In reality, we do not have the true model for $P(R = 1|X)$ but need to estimate it, typically by the logistic regression and other statistical models.

The inverse propensity w may be used to estimate a mean quantity of interest in the presence of incomplete data. In survey statistics typically an estimate of the population mean is calculated as a function of w and y_i from respondents, whereas in causal inference typically an average causal effect is calculated as a function of w and observed y_i whether in the treatment or control group.

In the survey context, a mean estimate of a survey outcome of interest can be computed from the respondents in a Horvitz-Thompson (HT)-type estimator (Horvitz and Thompson, 1952):

$$\frac{\sum_{i=1}^N w_i \times r_i \times y_i}{\sum_{i=1}^N w_i \times r_i}, \quad (3)$$

where, the r_i indicates whether the study unit responded to the survey ($r_i = 1$) or did not ($r_i = 0$). Note that the denominator of $\sum_{i=1}^N w_i \times r_i$ rather than n making the estimate a weighted average of respondents. The use of w reweights the respondents to mirror the full sample to achieve representative observed data. IPW was initially proposed in the survey literature as a means to adjust sample estimates to mirror the population by incorporating sample selection probabilities (Horvitz and Thompson, 1952), but naturally extends as a nonresponse bias adjustment.

In epidemiology and biostatistics, an average causal effect can be calculated Hernán and Robins (2020):

$$\frac{\sum_{i=1}^N w_i \times r_i \times y_i}{\sum_{i=1}^N w_i \times r_i} - \frac{\sum_{i=1}^N w_i \times (1 - r_i) \times y_i}{\sum_{i=1}^N w_i \times (1 - r_i)},$$

where the r_i indicates whether a study unit received an active treatment ($r_i = 1$) or a comparison control group ($r_i = 0$). In this estimator the counterfactual unobserved outcome for each study unit (e.g. y_i under treatment for a study unit with $r_i = 0$) is accounted for through the weighting the control group to mirror the treatment group using w .

2.2 An important condition for the correctness of IPW

Equation (1) explains that w aims to balance the distribution of X , i.e. w aims to weight and make the *multivariate* distribution $P(X|R = 1)$ approximate $P(X)$. Causal inference and missing data methodologies which use IPW are based on this notion (Hernán and Robins, 2020). Intuitively, since w balances with the multivariate distribution of X , the *univariate* distribution of each of p variables in X should be balanced accordingly. The following definition clarifies this intuition.

Definition 2. *The univariate balancing conditions for p covariates are defined in the form of the following estimating equations.*

$$\sum_{i=1}^N [w_i \times r_i \times (x_{ki} - \bar{x}_k)] = 0, \text{ for } k = 1, \dots, p. \tag{4}$$

where i denotes the i^{th} study unit, and $\bar{x}_k$ denotes the sample mean of X_k . The word *univariate* is used in comparison with *multivariate* covariate distributions because the latter, as shown in equation (1), allows balancing of complex interactions among covariates $X_1, X_2, \dots, X_p$ in addition to balancing each covariate separately. Correct IPW weights must certainly meet univariate balancing conditions (4) as written in the following remark.

Remark. *Let $\hat{\pi}(x_i)$ denote an estimated propensity score for study unit i , $\pi(x_i)$ denote the true propensity score, and $c(x_i)$ denote the ratio of $\pi(x_i)$ to $\hat{\pi}(x_i)$. If $c(x_i)=1$, which indicates the correctness of $\hat{\pi}(x_i)$, then the multivariate balancing condition (1) is a sufficient condition for univariate balancing conditions (4).*

Proof. Let w_i denote the estimated IPW weight for study unit i , which may or may not meet the multivariate balancing condition. The expectation with respect to the true response

probability $\pi(x_i)$ of the univariate balancing condition (4) for x_k is

$$\begin{aligned} \sum_{i=1}^N E[w_i \times R_i \times (x_{ki} - \bar{x}_k) | x_i] &= \sum_{i=1}^N E\left[\left(\frac{R_i}{\hat{\pi}_i} + \frac{1 - R_i}{1 - \hat{\pi}_i}\right) \times R_i \times (x_{ki} - \bar{x}_k) | x_i\right] \\ &= \sum_{i=1}^N E\left[\frac{R_i}{\hat{\pi}_i} \times (x_{ki} - \bar{x}_k) | x_i\right] = \sum_{i=1}^N \left\{ \frac{E[R_i | x_i]}{\hat{\pi}_i} \times (x_{ki} - \bar{x}_k) \right\} \\ &= \sum_{i=1}^n \left\{ \frac{\pi_i}{\hat{\pi}_i} \times (x_{ki} - \bar{x}_k) \right\} = \sum_{i=1}^n \{c(x_i) \times (x_{ki} - \bar{x}_k)\}. \end{aligned}$$

If w_i meets the multivariate balancing condition (1) then $c(x_i) = 1$ indicating a correctly estimated propensity, and the expectation of the univariate balance condition is equal to zero because $\sum_{i=1}^n (x_{ki} - \bar{x}_k) = 0$. □

This remark tells that correct IPW weights, created by the multivariate condition (1), must *always* meet the univariate balancing conditions (4). In other words, multivariate balance implies univariate balance; therefore, lack of univariate balance implies lack of multivariate balance. Thus, the correctness of IPW weights needs to be checked with the univariate balancing conditions. Such checking has been emphasized in major causal inference research works (Austin, 2011; Imai and Ratkovic, 2014; Morgan and Winship, 2015).

Even though balancing the multivariate distribution of X is a principal task of IPW weights, it is based on a response propensity model $P(R = 1|X)$, which is a single scalar summary of X . This requires IPW weights to have a correctly specified response propensity score. In the following section, a careful modeling task for the propensity score with a machine learning method is described.

2.3 Generalized boosted regression models (GBM) for IPW

The IPW weights w heavily rely on the correctness of the propensity model for estimating p . Response propensity is often estimated using logistic regression model, requiring direct specification of covariates and appropriate transformations (e.g. interaction terms) and assumes a specific distributional and functional form. Machine learning classification algorithms offer an alternative for determining a probability of response/selection. Specifically, generalized boosted regression models (GBM), also known as gradient boosting machines, is a machine learning algorithm that ensembles weak (classification) tree models sequentially in a non-parametric way (Friedman, 2001; Hastie et al., 2001).

GBM is implemented in the `gbm` package in R (Ridgeway, 2020; Greenwell et al., 2022). The `gbm` function requires selecting the number of iterations/trees T , the (interaction) depth of each tree K , and the shrinkage or learning rate parameter λ , along with a subsampling rate s . The algorithm first initializes $\hat{\pi}(x)$ to be the constant $\arg \min_{\rho} \sum_{i=1}^n \Psi(y_i, \rho)$, where the loss

function $\Psi(\cdot)$ is chosen to be the binomial deviance for the classification case of estimating propensity $\pi(x_i)$. Then for each iteration/tree t in $1, \dots, T$:

1. compute the negative gradient to use as the working response $z_i = -\frac{\partial}{\partial \pi(x_i)} \Psi(y_i, \pi(x_i))$
 $\big|_{\pi(x_i)=\hat{\pi}(x_i)}$;
2. randomly select $s \times n$ cases from the dataset;
3. using the randomly selected cases, fit a regression tree with K terminal nodes predicting z_i from the covariates x_i ;
4. compute the optimal terminal node predictions $\rho = (\rho_1, \dots, \rho_K)$ as $\rho_k = \arg \min_{\rho} \sum_{x_i \in S_k} \Psi(y_i, \hat{\pi}(x_i) + \rho)$, where S_k is the set of x_i that define terminal node k ; and
5. update the estimate of $\pi(x)$ as $\hat{\pi}(x_i) \leftarrow \hat{\pi}(x_i) + \lambda \rho_{k(x_i)}$, where $k(x_i)$ is the index of terminal node for observation i . (Ridgeway, 2020)

The $\hat{\pi}(x_i)$ after the last iteration/tree is the estimated value of the propensity score $P(R = 1|X = x_i)$. The last step in the algorithm describes a nonparametric specification of the propensity score by the additive element of $\hat{\pi}(x_i) + \lambda \rho_{k(x_i)}$. In particular, GBM does not require an assumption about a parametric form that the logistic regression model requires for the association between R and X . Furthermore, GBM can naturally handle high dimensionality, colinearity, nonlinearity, missing data, and complex interactions among covariates. Even though the ensemble nature of GBM makes the association between R and X not interpretable, there are measures of relative covariate importance to inform the impact of each of the covariates.

GBM is a general procedure for classification, typically with the purpose of achieving the highest predictive accuracy possible. The GBM tuning parameters (e.g. number of iterations/trees, the shrinkage rate, and tree interaction depth) allow control of the prediction bias-variance tradeoff through the amount of over-/under-fitting. In the context of adjusting for nonresponse/selection bias, however, the ultimate goal is covariate balance between the two study groups, not obtaining the optimal prediction of response/selection. Thus traditional use of GBM requires slight adaptations for the purpose of survey and causal analysis. The `twang` package in R builds upon the `gbm` function to fit GBM and calculates evaluation measures that are commonly used in causal analysis (Cefalu et al., 2021). The `ps` function in `twang` builds GBM-based response propensity models with automated incorporation of survey weights, and automated calculation of, for example, average treatment effects. Additionally, selection of the number of trees tuning parameter is carried out by optimizing covariate balance measures (e.g. via a Kolmogorov-Smirnov statistic) rather than through a measure of prediction error. Even though the package was designed for IPW for causal inference, its setup is easily adaptable to IPW for survey nonresponse.

3 Review of calibration weighting

3.1 The calibration formulation

Suppose we have the same data structure as in Table 1, but in addition we have target population means $\mu_k = E(X_k)$ for all $k = 1, \dots, p$ variables known from external sources. Classic survey calibration enables estimates of survey outcomes to be calculated from survey respondent data by using known target values of covariates X to adjust weights associated with survey respondents. Generalized raking calibration aims to create w_C to meet the following balancing conditions exactly or approximately (Deville and Särndal, 1992):

$$\frac{1}{N} \sum_{i=1}^N \left[w_{C,i} \times r_i \times (1, x_{1i}, \dots, x_{pi}) \right] = (1, \mu_1, \dots, \mu_p), \quad (5)$$

where $(\mu_1, \dots, \mu_p)$ is the p -dimensional vector of known target means making for p different univariate constraints in the calibration procedure. The generalized raking calibration formulation is typically described as calibrating to known target totals, but with an assumed fixed population size and fixed number of respondents n , calibration to means and totals are considered one in the same. These constraints are formulated together with an optimization to minimize the difference between initial survey inclusion base weights and calibration adjusted weights. The general formulation of this calibration optimization is

$$\min_{w_C} \sum_{i=1}^N \left[r_i \times w_{0,i} \times G(w_{C,i}/w_{0,i}) \right] \quad (6)$$

subject to the constraints equation 5, where G is a function selected to describe the discrepancy between the adjusted weights w_C and the original weights w_0 . Almost all of the classic survey calibration techniques (e.g. raking, ratio adjusting) can be fit into this generalized raking formulation via different choices of the distance metric G (Deville and Särndal, 1992; Wu and Lu, 2016; Slud and Morris, 2022). This adjusted weight w_C can then be used in an HT-type estimator as for IPW weights. Note that we do not incorporate initial inclusion base weights in this paper, i.e. we assume $w_{0,i} = 1$, so as not to over-complicate the simulated data.

3.2 Calibration weighting with working IPW weights

Recall that the weight w obtained from IPW is expected to hold the univariate balancing equality in equation (4). If this does not hold, then the response propensity is considered incorrectly specified; however calibration can be implemented for w to impose the equality. Calibration methods can be used to create an additional set of weights to enable the balancing equality.

If the working IPW weights do not meet the univariate balance conditions (4) due to propensity model misspecification, they can be further adjusted with calibration to impose the

conditions. Consider w_I as working IPW weights that violate univariate balancing:

$$E(w_I \times R \times X_k) \neq E(X_k), \text{ for some } k. \quad (7)$$

Implementing calibration introduces an additional factor w_C to enable final weights to meet (4) as seen below:

$$E(w_C \times w_I \times R \times X_k) = E(X_k), \text{ for the same } k \text{ in equation (7)}. \quad (8)$$

The additional factor w_C can be readily computed with any calibration technique. Let w^* denote the product of the additional calibration factor w_C and the working IPW weights w_I that fail to meet the univariate balancing conditions (4). Then we can use the calibration formulation to adjust the working IPW weights w_I :

$$\min_{w^*} \sum_{i=1}^N \left[r_i \times w_{I,i} \times G(w_i^*/w_{I,i}) \right] \quad (9)$$

subject to

$$\frac{1}{N} \sum_{i=1}^N \left[w_i^* \times r_i \times (1, x_{1i}, \dots, x_{pi}) \right] = (1, \mu_1, \dots, \mu_p). \quad (10)$$

In this calibration formulation, $w_{C,i}$ is precisely $w_i^*/w_{I,i}$ so that the additional factor w_C is created to ensure that $w^* = w_I \times w_C$. This notion implies that any working IPW weights can be coerced to meet the univariate balancing conditions by the calibration method.

3.3 Entropy balancing weighting (EBW) for calibrated IPW

The EBW method has been used extensively as a calibration weighting method for causal inference in political science (Hainmueller, 2012). It fits the general formulation of generalized raking calibration in equations 5 and 6 assuming the entropy distance metric $G_{entropy}(z) = z \log z$. For calibrated IPW this G function assumption reduces the optimization objective function for final weights in equation (9) to

$$\min_{w^*} \sum_{i=1}^N \left[r_i \times w_i^* \times \log(w_i^*/w_{I,i}) \right]. \quad (11)$$

We focus on EBW for calibration in this paper both on its own and together with IPW. Note that for the case of all categorical X variables, this formulation is equivalent to that of raking (Slud and Morris, 2022).

4 Simulation study: Revisiting Kang and Schafer (2007)

Kang and Schafer (2007) designed a simulation study to describe, evaluate, and compare competing methodologies for causal inference and missing data analysis; specifically outcome-based modeling with imputation, propensity-based modeling with weight-adjustment where propensity is estimated using logistic regression, and the combination of the two (doubly robust procedures). This well-cited study delves into the theoretical construct of double robustness, revealing interesting insight about the studied methods along the way. For example, Kang and Schafer (2007) note the sensitivity of propensity-based methods to model misspecification, as well as their generally inferior performance as compared to the doubly robust procedures. The study concluded that weight-adjustment methods, even in combination with outcome-based methods, were outperformed by outcome-based methods alone. We extend the set of propensity-based methods assessed in the Kang and Schafer (2007) simulation to describe, evaluate, and compare calibration (EBW), IPW with propensity estimated using GBM, and the combination of the two (calibrated IPW).

The simulation setup considers $N = 1000$ study units with variables z_{ki} for $k = 1, \dots, 4$ and noise ϵ_i sampled from standard normal distributions. The outcome y_i and propensity score π_i are defined as

$$y_i = 210 + 27.4z_{1i} + 13.7z_{2i} + 13.7z_{3i} + 13.7z_{4i} + \epsilon_i, \quad (12)$$

$$\pi_i = \pi(z_i) = 1 - (1 + \exp(-z_{1i} + 0.5z_{2i} - 0.25z_{3i} - 0.1z_{4i}))^{-1}.$$

The outcome y_i and $\text{logit}(\pi_i)$ have a direct linear relationship with $z_i = (z_{1i}, \dots, z_{4i})$. To introduce more complex relationships, Kang and Schafer (2007) define a set $x_i = (x_{1i}, \dots, x_{4i})$ formulated as combinations and transformations of variables z_i : $x_{1i} = \exp(z_{1i}/2)$, $x_{2i} = z_{2i}/(1 + \exp(z_{1i})) + 10$, and $x_3 = (z_{1i}z_{3i}/25 + 0.6)^3$, $x_4 = (z_{2i} + z_{4i} + 20)^2$. The x_i are made available for use in statistical models, whereas the z_i are not available. With the z_i unavailable for modeling it is unlikely that the true complex model formulation of y_i and π_i with x_i will be correctly specified, but more flexible techniques may be able to better capture the complex relationships. Study units that have a true π_i greater than a random uniform draw are assumed to have observed values of y_i , i.e. $r_i = 1$, whereas the others are assumed to have unobserved y_i . The simulation study goal is to estimate $E(Y)=210$ using information available for analysis: r_i , x_i , and y_i for $r_i = 1$.

In the subsequent sections we compare several weight-adjustment approaches: EBW calibration alone (C-EBW), IPW alone using GBM with interaction depth 1 (IPW-GBM1), IPW alone using GBM with interaction depth 2 (IPW-GBM2), calibrated IPW using GBM with interaction depth 1 (CIPW-GBM1), calibrated IPW using GBM with interaction depth 2 (CIPW-GBM2), and IPW along with logistic regression (IPW-GLM). Each of the methods yields estimated $P(R = 1|X)$ that translate to the sets of w_i . These sets of w_i are then used to determine HT-type weighted means (see equation 3) for the y_i , x_i , and z_i variables.

The two IPW-GBM methods estimate the response propensity $P(R = 1|X)$ using GBM with different values of interaction depth. Interaction depth, i.e. the number of splits, controls the size/complexity of the tree grown at each GBM iteration. A depth of one means that all trees can have at most a single split on a single variable resulting in two terminal nodes. A depth of two means that there can be at most two splits, therefore involving at most two variables, resulting in a two-way interaction. The IPW-GLM estimates the response propensity using logistic regression. The EBW method calibrates on all x_i to their complete data estimated mean one at a time, whereas the CIPW-GBM methods implement the EBW calibration on all x_i one at a time to their complete data estimated mean after an initial IPW weight adjustment using GBM.

4.1 Single simulation data analysis results

Table 2 presents results of outcome variable bias and covariate balancing based on the various weight-adjustment methods on a single data set generated from the Kang and Schafer (2007) simulation setup. The mean estimate of the outcome y_i in the last column of Table 2 is calculated as the weighted mean of y_i for study units with $r_i = 1$ (i.e. the HT-type estimator in equation 3) where the weight w_i is generated by each of the seven weighting methods. Similarly, the mean of x_k presented in Table 2 is the weighted mean of x_{ki} for study units with $r_i = 1$. That is, all results presented, except for the first row of true values, are weighted means calculated only from study units with $r_i = 1$.

Table 2: Estimated weighted mean for x_i and y_i for a single simulated dataset

Weighting Method	$\bar{x}_1$	$\bar{x}_2$	$\bar{x}_3$	$\bar{x}_4$	$E(Y)$
Truth	1.136	9.992	0.222	402.210	209.877
C-EBW	1.136	9.992	0.222	402.210	207.509
IPW-GBM1	1.086	10.014	0.219	403.602	206.634
IPW-GBM2	1.067	10.018	0.219	403.092	205.930
CIPW-GBM1	1.136	9.992	0.222	402.210	208.301
CIPW-GBM2	1.136	9.992	0.222	402.210	208.698
IPW-GLM	1.205	10.016	0.218	406.112	213.833

In these results we find that CIPW-GBM2 has the least biased estimate of the outcome mean: 208.7 compared to the true mean of 209.9 in this simulated dataset. As compared to the CIPW-GBM2 method, CIPW-GBM1 has a very slightly larger bias indicating a small effect of the interaction depth choice when using GBM to estimate propensity. The bias for the EBW alone (C-EBW) is slightly larger than the CIPW-GBM methods. All three IPW alone methods have a larger bias than the calibration alone and calibrated IPW approaches, regardless of the use of GBM or logistic regression to estimate propensity.

We can assess the notion of covariate balance in this simulated data by comparing the weighted means $\bar{x}_k$ calculated with only $r_i = 1$ study units to the overall observed mean

of x_i calculated on study units regardless of their r_i value. We find that the IPW alone methods have slightly less calibrated results on the covariates than the calibration alone and calibrated IPW methods. This is to be expected because the purpose of calibration is explicitly covariate balancing. In fact the methods with calibration have perfect covariate balancing, i.e. 1.14, 9.99, 0.22, and 402.2 for $\bar{x}_1$, $\bar{x}_2$, $\bar{x}_3$, and $\bar{x}_4$, respectively. The IPW alone methods display reasonable covariate balance despite that not being the explicit goal of IPW.

Looking at outcome bias and covariate balancing together, we find that the calibrated IPW methods perform best on both measures. This result suggests that there is statistical value in incorporating multivariate relationships with IPW, but not without an additional “correction” for covariate balancing in the univariate dimension. Recall that calibration aims to attain the p univariate balancing conditions (4): $E(w \times R \times X_k) = E(X_k)$ for all $k=1, \dots, p$, which does not allow interactions among the covariates, whereas IPW aims to attain the multivariate balancing property $w \times P(X_k | R = 1) = P(X_k)$, which allows complex interactions among the covariates.

Recall that x_i , not z_i , are available for statistical modeling; that is, x_i are used as covariates in the propensity models, and x_i are used as targets in the calibration. The z_i , however, are available in the simulation setup. Table 3 presents measures of variable balance on the z_i . The first four columns of results in the table displays the difference between weighted means $\bar{z}_k$ calculated with only $r_i = 1$ study units and the overall observed mean of z_i calculated on study units regardless of their r_i value. The last column in the table calculates the sum of the absolute value of the z_i differences across the row, representing an overall measure of balance in z_i .

Table 3: Difference between true sample mean and weighted estimated mean of z_i for a single simulated dataset

Weighting Method	z_1	z_2	z_3	z_4	Overall
C-EBW	-0.016	0.119	-0.142	-0.117	0.394
IPW-GBM1	-0.062	0.089	-0.144	-0.057	0.352
IPW-GBM2	-0.092	0.081	-0.126	-0.058	0.357
CIPW-GBM1	0.021	0.065	-0.152	-0.068	0.306
CIPW-GBM2	0.023	0.055	-0.131	-0.055	0.264

We find that the overall z_i -balance is best for the CIPW-GBM2 method; that is, the measure of overall difference is lowest (0.264). As compared to the CIPW-GBM2 method, CIPW-GBM1 has a slightly larger overall difference indicating an effect of the interaction depth parameter in GBM. Recall an increase in GBM model complexity (i.e. a larger interaction depth value) increases the ability to capture complex interactions. Our results suggest that a more flexible GBM better models the complex relation between r_i and z_i . The z_i -balance is worse for the IPW alone methods and worst for the calibration along method (C-EBW).

These results contrast with those measures of balance with respect to x_i in Table 2, where the methods using calibration obtained the best – in fact perfect – x_i -balance, by construct. This suggests that the CIPW (and less so the IPW) methods have reasonably modeled the association between z_i and r_i in handling complex interactions among the covariates even though the univariate balance conditions are not perfectly met. The weights obtained from the IPW methods were improved by the EBW step as implemented in CIPW; that is, the calibration on x_i following IPW better balanced the unobserved-to-the-model z_i . This finding illustrates a complicated balance between capturing complex relationships between r_i and covariates, and meeting univariate balancing conditions.

4.2 Repeated simulation study results

The results in the previous section describe the performance of the various weight adjustment methods on just one realized dataset of the simulation setup. This analysis is informative, but a study of multiple realizations of the simulation setup will detail the statistical properties of the various methods. In this section, the various weight adjustment methods are applied to 1000 simulated datasets in order to assess outcome estimation and covariate balance.

4.2.1 Outcome estimation

Kang and Schafer (2007) reported that in this repeated simulation study scenario, linear regression (OLS) imputation had the least bias in outcome estimation, as compared to IPW-GLM and doubly robust procedures. The OLS imputation model assessed is an outcome-based modeling approach, as opposed to the propensity-based modeling approaches assessed in Section 4.1. Specifically, the OLS imputation approach first fits a linear regression of y_i on x_i for study units with $r_i = 1$ to obtain regression coefficients $\hat{\beta}$. These regression coefficients are then used to impute y_i values for study units with $r_i = 0$ as $\frac{1}{N} \sum_{i=1}^N x_i^T \hat{\beta}$. Recall that there are important practical (in addition to statistical) differences between outcome-based and propensity-based methods: propensity-based weighting methods do not incorporate any information about y_i , but instead rely on the r_i and x_i relationship.

Table 4 shows measures of performance with respect to outcome mean estimation of all the weighting methods and the OLS imputation method. Letting $\hat{\mu}^{(b)}$ be the HT-type estimate for the b^{th} simulated sample, the measures in Table 4 are defined as:

$$\begin{aligned}
Bias^{(b)} &= 210 - \hat{\mu}^{(b)} \\
MeanBias &= \frac{1}{10^3} \sum_{b=1}^{1000} Bias^{(b)} \\
VarBias &= \frac{1}{10^3 - 1} \sum_{b=1}^{1000} (Bias^{(b)} - MeanBias)^2 \\
RMSE &= \sqrt{\frac{1}{10^3} \sum_{b=1}^{1000} (Bias^{(b)})^2} \\
MAE &= \text{median}(|Bias^{(1)}|, \dots, |Bias^{(10^3)}|) \\
RNG &= 2.5 \text{ to } 97.5 \text{ percentile of } (Bias^{(1)}, \dots, Bias^{(10^3)}).
\end{aligned}$$

We find that CIPW-GBM2 outperforms the OLS method in the variability of the bias (Var-Bias), RMSE and MAE criteria. The range for the CIPW-GBM2 method is also narrower than for the OLS method. Recall that Kang and Schafer (2007) determined OLS as the superior method among their comparison methods. Our result places the more flexible and robust calibrated IPW method as a direct competitor to OLS, whereas IPW alone absent univariate covariate balance is not enough as suggested in the discussion of Kang and Schafer (2007). Results in Table 4 indicate that machine learning-based IPW weights, which are also properly calibrated, are as good as the outcome-based OLS method.

The CIPW-GBM2 sees an advantage over CIPW-GBM1 across all performance measures. The 44% larger mean bias and 7% larger variability in the bias for CIPW-GBM1 suggest that the bias introduced by fitting a less complex propensity model (i.e. interaction depth one instead of two) is not compensated for with a large enough variance reduction. In other words, in this simulation study, the bias reduction from fitting a deeper propensity model is worth more than the variance reduction from simpler model when it is used in conjunction with calibration.

Looking at the calibration and IPW alone results, C-EBW exhibits a smaller mean bias, but larger variability in the bias than the IPW alone methods. Taking mean bias and bias variability together, the IPW alone methods have larger RMSE and MAE than C-EBW. The poorly performing IPW alone methods are greatly improved with the calibration step as implemented in the CIPW-GBM methods.

Table 4: Evaluation measures of outcome y_i estimation with 1000 datasets

Method	MeanBias	VarBias	RMSE	MAE	RNG	
C-EBW	1.881	2.310	2.418	1.894	-1.005	4.966
IPW-GBM1	2.997	1.806	3.285	2.973	0.360	5.666
IPW-GBM2	3.394	1.745	3.642	3.383	0.789	5.972
CIPW-GBM1	1.271	1.879	1.869	1.365	-1.342	4.024
CIPW-GBM2	0.880	1.754	1.590	1.098	-1.726	3.514
OLS	0.725	2.263	1.669	1.159	-2.213	3.693

4.2.2 Covariate balancing

Unbiased estimation of the outcome mean is the ultimate goal of weight adjustment methods, but the bias is not measurable in real data analysis because the true outcome mean is not observed. The covariates, however, are assumed observed for all study units and available to assess and compare between groups. Standardized difference (Austin, 2011) has been used to assess the performance of the balancing properties of the covariates. This measure is defined as

$$\frac{\bar{x}_{group1} - \bar{x}_{group2}}{\sqrt{s_{group1}^2 + s_{group2}^2}}, \tag{13}$$

where $\bar{x}_{group1}$ and $\bar{x}_{group2}$ denote the sample mean of the covariate for group one and group 2 subjects, respectively, whereas s_{group1}^2 and s_{group2}^2 denote the sample variance of the covariate in group one and group two subjects. In the original formulation of standardized difference in causal inference, group one is the treatment group and group two is the control group. In the survey context, group one is the respondent group and group two is the nonrespondent group.

The standardized difference in equation (13) is simple to formulate because the group one (e.g. treated or responding) subjects are independent from the group two (e.g. control or nonresponding) subjects. In the context of nonresponse adjustment, however, covariate comparisons between the respondents and the entire sample is desired. In this scenario of comparing the sample mean of the covariates between group one subjects ($R = 1$) and the entire set of subjects ($R = 0$ or $R = 1$), the two group sample means are correlated with each other. This correlation implies that the denominator in equation (13) is not a valid estimate of the standard deviation of the difference in means between the two overlapping groups. Bootstrap resampling can be used instead for estimating the variance of the difference in means. This is particularly convenient in the simulation study as 1000 random samples have already been drawn from the simulation study design (12). A resampling version of the standardized difference is defined as follows:

$$\frac{\bar{\Delta}_B}{\sqrt{\sum_{b=1}^B (\Delta_b - \bar{\Delta}_B)^2 / (B - 1)}},$$

where $\bar{\Delta}_B = \frac{1}{B} \sum_{b=1}^B \Delta_b$, $\Delta_b = (\bar{x}_{1b} - \bar{x}_b)$, $\bar{x}_{1b}$ indicates the sample mean of covariate in the respondent/treated ($R = 1$), and $\bar{x}_b$ is the sample mean of covariate for all units in the b^{th} simulation data set.

Table 5: Covariate balancing (standardized difference) of x_i with 1000 datasets

Weighting Method	$\bar{x}_1$	$\bar{x}_2$	$\bar{x}_3$	$\bar{x}_4$
C-EBW	0.394	0.530	0.050	0.443
IPW-GBM1	2.877	3.236	1.331	1.348
IPW-GBM2	4.165	3.894	1.051	1.189
CIPW-GBM1	0.440	0.438	0.265	0.426
CIPW-GBM2	0.031	0.032	0.032	0.045

Table 5 shows standardized differences for x_i calculated via the resampling version with $B = 1000$. It is commonly accepted that a standardized difference less than 0.1 indicates a negligible difference in the covariate means (Austin, 2011). We find that CIPW-GBM2 is the only method that is under the 0.1 threshold, and it is under that threshold for all four covariates. The CIPW-GBM1 and C-EBW methods have similar levels of covariate balance to each other, albeit with standardized differences that are quite larger than for CIPW-GBM2 and exceed the 0.1 threshold. The IPW alone methods exhibit the most violation in covariate balance.

The ranking of performance for the five methods based on covariate balancing are equivalent to the ranking of performance for the five methods based on outcome estimation bias. This suggests, at least in this simulation setup, that assessing covariate balance in this way between respondents/treated subjects and the entire sample signals the methods' ability to adjust nonresponse/selection bias.

5 Discussion

This paper reviews the developments of generalized boosting models for propensity estimation in IPW, and the combination of IPW and calibration weighting. The simulation study extends the set of possible methods beyond those studied in Kang and Schafer (2007) to include GBM-based IPW and the combined calibration and IPW method. It is a striking result that the calibrated IPW weights, which do not use the outcome information at all, outperforms the OLS imputation method, which uses the outcome information. Kang and Schafer (2007) concluded that OLS imputation outperforms all IPW approaches, but results

in this paper update this conclusion in the presence of a more complex propensity model and a further calibration adjustment. This suggests that when weights are built carefully to balance both uni- and multi-variate balancing conditions, weight adjustment methods can be as good as estimators from an outcome-based method.

Calibration for weight adjustment to address nonresponse/selection bias does not implicitly address interactions among covariates. For example, if there is a strong relationship between selection/nonreponse propensity and jointly X_1 and X_2 , then calibration requires the product of X_1 and X_2 to be included as a calibration variable explicitly. On the other hand, GBM-based IPW naturally allows for all such interactions without explicit pre-processing of covariates. The GBM-based IPW method, however, does not guarantee that the univariate balancing conditions (4) hold. The univariate balancing conditions should be empirically assessed after the GBM-based IPW weight adjustment, and further refined as necessary with a calibration adjustment to the GBM-based IPW weights.

Results in this paper are consistent with Mercer et al. (2018) regarding calibrated IPW. Simulation study evidence suggests that combining calibration and IPW generally provides better performance than IPW alone or calibration alone. This suggests that the use of machine learning models that naturally capture interactions (random forests in Mercer et al. (2018), GBM in this paper) provide value in estimating complex propensity relationships to achieve multivariate covariate balance, but can be improved by ensuring the univariate balancing conditions are met through calibration. This advantage may of course depend on the particular application and dataset: if propensity truly depends simply on main effects of the covariates, then using an overly complex model for IPW to achieve multivariate balancing conditions is likely unnecessary.

Comparisons in this paper assess the adaptation of IPW that uses GBM for propensity estimation. There are a plethora of machine learning methods for classification that could also serve as a flexible alternative to the traditionally used logistic regression. The focus on GBM is motivated by its use in previous comparisons (e.g. Lee et al. (2010)) and its popular implementation in causal inference via the `twang` package in R (Cefalu et al., 2021; Ridgeway, 2020). Comparisons in this paper regarding calibration only assessed entropy-based calibration weighting. There are a plethora of other calibration methods such as exponential tilting and iterative proportional fitting. The focus on EBW is motivated by its well-cited use in the causal inference literature (Hainmueller, 2012) and its associated implementation in the `ebal` package in R (Hainmueller, 2022). A complete comparison of all machine learning propensity models (including selection of tuning parameters via covariate balancing) and calibration techniques is not considered in the scope of this work.

There are many considerations for future research efforts in extending this study. First, the simulation setup assumes four variables, all of which had a defined relationship with both the outcome and the propensity. One may, however, have a much larger set of covariates of which some may not have strong relationships with the outcome and/or the propensity. The effect of covariate selection is an important and practical consideration. For the com-

bined IPW approach, the performance may also be affected by the selection of covariates to further balance with a calibration step. That is, if only a subset of covariates violate the univariate balance conditions after IPW, then a calibration on only those covariates may be necessary to achieve the benefits of the combined IPW method. Second, the ultimate focus in evaluation of the methods in this paper is outcome mean estimation; however complex modeling and unnecessary adjustment may result in increased variance. Third, this study does not address the use of sample selection weights in the context of survey analysis. There is precedent for incorporating complex survey designs into machine learning methods (e.g. GBM), but that specification is not assessed in this work. Lastly, in survey research, reliable estimates of demographic variables from a large national survey or a census are typically used as calibration targets (Slud and Morris, 2022). This is particularly important when calibration is used for benchmarking, as opposed to covariate balancing within a survey frame for nonresponse adjustment. The simulation study in this paper uses the complete data auxiliary variable's mean as the calibration targets for the EBW and calibrated IPW approach, in line with the causal inference literature. In the survey context, however, the use of estimates of target variable quantities rather than a strong source of auxiliary information is an important matter of future study.

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No figures in this paper

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